

# AI\_reference.pdf

## NTHU Chatroom Midterm Project - AI Usage Report

AI Tool(s) Used: ChatGPT (GPT-5.5 Thinking).

Scope of Usage: AI was used for debugging Firebase/Auth/Firestore issues, implementing requested chatroom features, writing README.md, and preparing this AI usage document. The final code was manually tested and adjusted during development.

### 1. Code segments assisted by AI

| Location                                                                                   | Prompt used                                                                                                                                                                                    | AI output / refinement                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| index.html, lines 8-110;<br>main.js, lines 188-246                                         | Initial prompt: The PPT is what I should do, the zip file is my chatroom project, and the screenshot is the error. Please solve it. Later prompts asked to fix sign in/up and Firebase errors. | AI helped restructure Firebase initialization, email/password authentication, Google sign-in button, and profile creation. I kept Firebase compat SDKs and used <code>___/firebase/init.js</code> for hosting config.                                           |
| main.js, lines 387-400;<br>index.html, lines 83-105;<br>style.css, profile-related classes | Prompt: Change the profile part so users can upload profile instead of using URL/base64, and display users profile in chatroom.                                                                | AI provided image upload and resize-to-data-URL logic, preview update, Firestore profile save, and message avatar rendering. I kept image data small to avoid Firestore field-size errors.                                                                      |
| main.js, lines 413-455 and<br>507-556                                                      | Prompt: Fix room creation, direct/group classification, and message loading behavior.                                                                                                          | AI suggested deterministic direct room IDs, room type direct/group, group invite update, realtime listeners, and scroll behavior. I refined the room UI text and direct/group naming.                                                                           |
| main.js, lines 558-590 and<br>697-755                                                      | Prompt: Fix sending messages, message rendering, avoid blinking after sending, and allow code to be sent safely.                                                                               | AI helped build <code>addMessage/sendMessage</code> and <code>renderMessages</code> with escaped HTML via <code>clean()/formatMessageText()</code> . I changed animations so only new messages receive the new-message class, avoiding full-page blinking.      |
| main.js, lines 606-646 and<br>image input handler around<br>lines 317-356                  | Prompt: Fix image sending when Firestore says imageData is longer than the allowed size.                                                                                                       | AI provided canvas-based image resize/compression before storing imageData. I used JPEG quality reduction and a size check to avoid exceeding Firestore limits.                                                                                                 |
| main.js, lines 777-822 and<br>emoji picker code around<br>lines 286-315, 660-695           | Prompt: Emoji reactions fail on image/self messages and the emoji picker does not open for my own messages.                                                                                    | AI fixed reactions by treating reactions as an array, using transactions for add/remove, and positioning the picker with fixed coordinates clamped inside the viewport.                                                                                         |
| main.js, lines 824-856                                                                     | Prompt: Implement Chrome notifications.                                                                                                                                                        | AI added Notification permission request and unread incoming message notifications only when the tab is not focused. I left notifications behind a user click because browsers require permission.                                                              |
| main.js, lines 80-139 and<br>879-944; firestore.rules, lines<br>24-61                      | Prompt: Implement block/unblock user according to the PPT: direct messages disabled with warning, group chat messages mutually hidden.                                                         | AI helped define direct-chat blocking via <code>room.blockedBy</code> , group filtering by <code>blockedUsers</code> , UI warning/disable logic, and Firestore rule enforcement for direct rooms. I refined unblock to update the deterministic direct room ID. |
| style.css, lines 1-172                                                                     | Prompt: Fix initial scroll/input box layout, add CSS animation, and improve UI.                                                                                                                | AI suggested CSS grid fixes: <code>height: 100dvh</code> , <code>minmax(0, 1fr)</code> , <code>min-height: 0</code> , scrollable messages, and animations for login, rooms, and new messages.                                                                   |
| README.md;<br>AI_reference.pdf                                                             | Prompt: Check whether everything except bonus is implemented; if so, write README.md and AI usage.                                                                                             | AI generated the implementation checklist, usage instructions, setup steps, testing checklist, and this report. I documented that the optional framework advanced item is not claimed.                                                                          |

## 2. Representative prompt and response records

### *Firestore initialization/auth*

**Prompt:** Prompt: The screenshot shows auth/invalid-api-key and later Firebase App named [DEFAULT] already exists. Please solve it.

**Response used:** AI response: Use Firebase Hosting `/__/firebase/init.js` for deployed config and guard `initializeApp` so Firebase is not initialized twice. Then use `firebase.auth()` and `firebase.firestore()`.

### *Firestore permissions*

**Prompt:** Prompt: After signing up, Missing or insufficient permissions appears and I cannot create rooms or send messages.

**Response used:** AI response: Update `firestore.rules` so authenticated users can read rooms where their uid is in `resource.data.members`, create messages only as sender, and later enforce direct-room block status.

### *Direct/group blocking*

**Prompt:** Prompt: Fix block so User B cannot send direct messages to User A anymore, history still shows warning, and group messages are mutually hidden.

**Response used:** AI response: Use type `direct` for two-person rooms and type `group` after inviting a third member. Store direct-room `blockedBy` list, disable input and show warning in direct chat, and filter messages in groups if either side blocked the other.

### *Safe code sending*

**Prompt:** Prompt: Implement deal with problems when sending code.

**Response used:** AI response: Escape `&`, `<`, `>`, quotes before inserting message text into `innerHTML` and preserve line breaks/spaces so scripts and HTML tags display as text instead of executing.

## 3. Refinement and explanation

- I rewrote and adjusted AI-provided snippets to match the project IDs and existing file structure, for example using `current-room-name`, `room-members`, `msg-input`, and `auth-container` instead of earlier mismatched IDs.
- I tested and fixed Firebase errors iteratively: invalid API key, duplicate initialization, permission errors, blocked queries, image field-size errors, and stale listeners after logout.
- The final code uses vanilla JavaScript rather than a framework. Therefore the optional framework advanced component is not claimed.
- The submitted ZIP should exclude `node_modules`, `.git`, `.firebase` cache, swap files, and backup files.

## 4. Statement of non-usage

This assignment did use AI tools, so the statement "No AI tools were used in this assignment" does not apply.